

Financial Statement Analysis

Introduction

Accuracy of the books of accounts is determined by means of preparing a Trial Balance. Having determined the accuracy of the books of accounts every businessman is interested in knowing about two more facts. They are: (i) Whether he has earned a profit or suffered a loss during the period covered by the Trial Balance, (ii) Where does he stand now? In other words, what is his financial position? The determination of the Profit or Loss is done by preparing a Trading and Profit and Loss Account (or an Income Statement). The financial position is judged by means of preparing a Balance Sheet of the business. The two statements together, i.e., Income Statement and the Balance Sheet are termed as Final Accounts. As the term indicates, Final Accounts means accounts which are prepared at the final stage to give the financial position of the business. In the present unit we are dealing with the basic principles concerning financial reporting particularly with reference to a non-corporate entity i.e. a sole proprietary or a partnership firm. Financial statements are the **final output of the accounting cycle**. They summarize the financial activities of a business and provide information about its financial health. For companies, financial statements are also a **legal requirement** and must comply with the **Companies Act, 2013** and relevant **Ind AS (Indian Accounting Standards)**.

Key Points:

- Companies are subject to higher disclosure requirements due to their use of significant societal resources.
- Stakeholders such as investors, creditors, employees, and regulators rely on these statements to make informed decisions.
- Schedule III of the Companies Act, 2013, prescribes the formats of the Balance Sheet and Statement of Profit and Loss for standardization.
- Ind AS 1 provides the framework for **presentation and comparability** with previous periods and other entities.

Purpose of Various Financial Statements

Financial statements provide detailed information for **decision-making, performance evaluation, and planning**. Each statement serves a distinct purpose:

1 Statement of Profit and Loss (Income Statement)

Purpose:

- Measures a company's **financial performance** over a period.
- Summarizes Income earned and expenses incurred to calculate **net profit or net loss**.
- Evaluates operational efficiency and profitability.
- Assists management in **cost control and revenue planning**.
- Provides investors information on **return on investment**.
- Helps creditors assess the **repayment capacity**.

Format of Income Statement (Vertical as per Schedule III):

Particulars	Amount (₹)
Revenue from Operations	xxx
Other Income	xxx
Total Income	xxx

Particulars	Amount (₹)
Expenses:	
- Cost of Materials Consumed	xxx
- Employee Benefits Expense	xxx
- Finance Costs	xxx
- Depreciation & Amortization	xxx
- Other Expenses	xxx
Total Expenses	xxx
Profit before Tax (PBT)	xxx
Less: Tax Expense	xxx
Net Profit after Tax (PAT)	xxx
Other Comprehensive Income (OCI)	xxx
Total Comprehensive Income	xxx

Recognition Principles:

- Accrual basis: Expenses and Income recorded when incurred or earned.
- Income recognized if probable and reliably measurable.

Example:

Revenue: ₹10,00,000; Expenses: ₹7,00,000 → Net Profit: ₹3,00,000

Balance Sheet (Statement of Financial Position)

Purpose:

- Provides a **snapshot of financial position** at a specific date.
- Shows **assets, liabilities, and equity**.

Accounting Equation:

Assets = Liabilities + Equity

Format (Vertical as per Schedule III):

Assets:

Particulars	Amount (₹)
Non-Current Assets:	
- Property, Plant & Equipment	xxx
- Intangible Assets	xxx
- Long-term Investments	xxx

Particulars	Amount (₹)
- Other Non-current Assets	xxx
Total Non-Current Assets	xxx
Current Assets:	
- Inventories	xxx
- Trade Receivables	xxx
- Cash & Cash Equivalents	xxx
- Short-term Loans & Advances	xxx
Total Current Assets	xxx
Total Assets	xxx
Equity & Liabilities:	
Particulars	Amount (₹)
Equity:	
- Share Capital	xxx
- Reserves & Surplus	xxx
Total Equity	xxx
Non-Current Liabilities:	
- Long-term Borrowings	xxx
- Deferred Tax Liabilities	xxx
Total Non-Current Liabilities	xxx
Current Liabilities:	
- Short-term Borrowings	xxx
- Trade Payables	xxx
- Current Tax Liabilities	xxx
- Other Current Liabilities	xxx
Total Current Liabilities	xxx
Total Equity & Liabilities	xxx

Financial statements provide financial information about a business enterprise to meet the information needs of decision-makers. This guide explains how to analyze, compare, and interpret financial data for informed decision-making.

Financial Statement Analysis is the process of examining, understanding, and interpreting a company's financial statements—**Income Statement, Balance Sheet, and Cash Flow Statement**—to evaluate its performance and financial health.

Purpose of Financial Statement Analysis

1. **Assess Profitability** - Is the business earning profit?
2. **Evaluate Liquidity** - Can the business pay its short-term bills?
3. **Understand Solvency** - Can the business survive long-term? Does it have too much debt?
4. **Check Efficiency** - How well is the business using its assets (inventory, machines, cash)?
5. **Support Decision-Making**
 - Investors: Should we invest?
 - Banks: Should we give loans?
 - Managers: How to improve performance?

Users of Financial Statement Analysis

- Management
- Investors
- Banks
- Suppliers and creditors
- Government agencies
- Employees

Methods of Analysis

1. **Trend Analysis**
2. **Comparative Analysis**
3. **Common-Size Analysis**
4. **Ratio Analysis**

2. Comparative Analysis

Comparative Analysis compares financial statements of **two or more years** to identify changes in financial performance.

It answers: "**What has changed from last year to this year?**"

You prepare a statement showing:

- Figures for Year 1
- Figures for Year 2
- Absolute change (Year 2 - Year 1)
- Percentage change

Why Comparative Analysis is Important

- Shows **growth or decline**
- Helps identify **trends**
- Highlights **strengths and weaknesses**
- Useful for forecasting future performance
- Helps managers take corrective action

3. How Ratio Analysis Helps in Comparative Analysis

Comparative analysis means comparing financial performance across two or more years OR comparing two companies.

But raw numbers alone cannot give the full picture because:

- One year may have higher revenue from operations simply due to inflation
- One company may be bigger than the other
- Expenses may increase but may still be efficient

This is why Ratios are used. Ratios **standardize** financial data and convert numbers into meaningful relationships.

Examples of Comparison

A) Comparing Same Company's Performance over years

Particulars	2023	2024	Change	% Change
Revenue from operations	₹10,00,000	₹12,00,000	+₹2,00,000	+20%
Net Profit	₹1,00,000	₹1,50,000	+₹50,000	+50%

B) Comparing Two Companies

Example: D-Mart vs Big Bazaar

Ratio	D-Mart	Big Bazaar	Who's Better
GPM	15%	22%	Big Bazaar (higher margin)
NPM	7%	2%	D-Mart (more efficient costs)
Inventory Turnover	20x	6x	D-Mart (faster stock movement)
D/E	0.1	1.3	D-Mart (less risk)

C) Comparing with Industry Standards

Example:

- Industry Avg Current Ratio = 1.5
- Company Current Ratio = 0.9
- **Interpretation** → liquidity risk

4. Accounting Ratios

Meaning

An **accounting ratio** is a mathematical number calculated as a reference to the relationship of two or more numbers from financial statements. It can be expressed as a fraction, proportion, percentage, or number of times.

Objectives of Ratio Analysis

1. Identify areas of business needing attention
2. Identify potential areas for improvement
3. Provide deeper analysis of profitability, liquidity, solvency, and efficiency
4. Enable cross-sectional analysis with industry standards
5. Provide information for projections and estimates

Advantages of Ratio Analysis

1. **Helps understand efficacy of decisions** - Indicates how decisions have improved performance
2. **Simplifies complex figures** - Establishes relationships and summarizes information
3. **Helpful in comparative analysis** - Reveals trends over multiple years
4. **Identifies problem areas** - Highlights areas needing attention
5. **Enables SWOT analysis** - Helps understand threats, opportunities, strengths, and weaknesses
6. **Various comparisons** - Allows intra-firm, inter-firm, and industry comparisons

Limitations of Ratio Analysis

From Financial Statements:

1. **Limitations of accounting data** - Based on conventions and judgments
2. **Ignores price-level changes** - Inflation affects comparability
3. **Ignores qualitative aspects** - Only monetary factors considered
4. **Variations in accounting practices** - Different policies affect comparability
5. **Forecasting limitations** - Historical data alone insufficient

From Ratios Themselves:

1. **Means, not the end** - Ratios are indicators, not solutions
2. **Lack of standardized definitions** - No universal definitions
3. **Lack of universal standards** - No ideal ratio levels
4. **Based on unrelated figures** - Some ratios may be meaningless

Types of Ratios

Functional Classification

1. **Profitability Ratios** - Analyze profits in relation to revenue from operations or assets
2. **Activity/Turnover Ratios** - Measure efficiency of operations
3. **Liquidity Ratios** - Measure ability to meet short-term obligations
4. **Solvency Ratios** - Measure ability to meet long-term obligations

Profitability Ratios

1. Gross Profit Ratio

Formula:

Gross Profit Ratio = $(\text{Gross Profit} / \text{Net Revenue from Operations}) \times 100$

Significance: Indicates gross margin on products sold and margin available to cover expenses.
Higher ratio is favorable.

Interpretation:

Shows production/operational efficiency.

Example:

A fall from **35% to 28%**

→ Raw material costs increased, or selling price reduced.

2. Net Profit Ratio

Formula:

Net Profit Ratio = $(\text{Net Profit} / \text{Revenue from Operations}) \times 100$

Generally, net profit refers to Profit after Tax (PAT).

Significance: Measures net profit margin and reflects overall efficiency.

Interpretation:

- Higher margin = better cost control and pricing power.
- Compare with competitors and past trends.

Example:

If NPM increases from **8% to 12%**,
→ Improved efficiency or better pricing.

3. Return on Capital Employed (ROCE)**Formula:**

$\text{ROCE} = (\text{Profit before Interest and Tax} / \text{Capital Employed}) \times 100$

Significance: Measures return on long-term funds employed in the business.

4. Return on Shareholders' Funds (RONW)**Formula:**

$\text{RONW} = (\text{Profit after Tax} / \text{Shareholders' Funds}) \times 100$

Significance: Assesses whether investment generates reasonable return for shareholders.

Interpretation:

- Measures how efficiently the company is generating profits from its **long-term funds** (equity + long-term debt).
- Higher ROCE indicates better utilisation of capital.
- Should always be compared with:
 - (a) Industry average
 - (b) Company's past performance
 - (c) **Cost of Capital** → the minimum return required by lenders and shareholders.

Important:

A business **must earn a ROCE higher than its overall Cost of Capital.**

- Cost of Capital includes:
 - ✓ Cost of Debt (interest rate)
 - ✓ Cost of Equity (expected by shareholders)

If ROCE < Cost of Capital → the business is destroying value.

If ROCE > Cost of Capital → the business is creating value.

- Suppose the company's banks charge **10% interest**, and shareholders expect a minimum **12%** return.
- Overall Cost of Capital $\approx$ **10–12%** (approx.).
- Then ROCE should be **more than 12%** for the business to be considered efficient.

Example:

- ROCE of Company A = **8%**
→ Below cost of capital (10–12%) → *Not efficient; value destruction.*
- ROCE of Company B = **15%**
→ Above cost of capital → *Value creation; efficient use of long-term funds.*

Interpretation:

- Measures the return the company is generating **for its owners (equity shareholders)**.
- Higher ROE indicates better utilisation of shareholders' capital.
- Should always be compared with:
 - (a) Industry average
 - (b) Company's own past performance
 - (c) **Sure return available to investors (e.g., bank FD rate $\approx$ 7%)**

Important:

If a safe bank deposit offers **~7%**, a business should generate **more than 7% ROE** to justify the higher risk taken by shareholders.

If ROE is below this “risk-free benchmark,” it suggests that the shareholders might be better off investing money in low-risk instruments rather than in the company.

Example:

- ROE of Company A = **5%** → *Lower than 7% bank return → poor for shareholders*
- ROE of Company B = **14%** → *Higher than 7% → good return and justifies risk*

Liquidity Ratios

1. Current Ratio

Formula:

Current Ratio = Current Assets / Current Liabilities

Components:

- **Current Assets:** Inventories, trade receivables, cash, short-term loans and advances
- **Current Liabilities:** Trade payables, short-term borrowings, short-term provisions

Significance: Measures degree to which current assets cover current liabilities. Provides safety margin. Should be reasonable—neither too high nor too low.

Interpretation:

- 1 means the firm can cover its short-term liabilities.
- 2 Too high (>2.5) may indicate idle assets.
- 3 Too low (<1) signals possible liquidity stress.

Example:

If Current Ratio = **0.62** (Tata Steel's approx value),

→ The company has only ₹0.62 of current assets for every ₹1 of liabilities → **tight liquidity position**

2. Quick (Acid-Test) Ratio

Formula:

Quick Ratio = Quick Assets / Current Liabilities

Where: Quick Assets = Current Assets - (Inventories + Advance tax)

Significance: Better measure than current ratio as it excludes less liquid assets.

Interpretation:

- Focuses on **most liquid** assets.
- A very low quick ratio means the firm depends heavily on selling inventory to pay bills.

Example:

Quick Ratio = **0.20**

→ Only ₹0.20 of quick assets available for ₹1 of liabilities → **high short-term liquidity risk.**

Solvency Ratios

1. Debt-Equity Ratio

Formula:

Debt-Equity Ratio = Long-term Debts / Shareholders' Funds

Where:

- Shareholders' Funds = Share Capital + Reserves & Surplus + Money received against share warrants + Share application money pending allotment

Significance: Measures degree of indebtedness. Low ratio reflects more security. Normally, 2:1 is considered safe.

Interpretation:

2 = High financial risk

<1 = Safer capital structure

Depends on industry (e.g., utilities run safely with high debt)

Example:

D/E = 0.45

→ Company is conservatively financed lower long-term risk.

2. Debt to Capital Employed Ratio

Formula:

Debt to Capital Employed Ratio = Long-term Debt / Capital Employed

Where: Capital Employed = Long-term Debt + Shareholders' Funds

Where:

- **Long-term Debt** = long-term borrowings, debentures, term loans
- **Capital Employed** = Shareholders' Funds + Long-term Debt

Significance: Shows proportion of long-term debts in capital employed.

Interpretation:

This ratio shows how much of the company's long-term capital structure is financed through **debt**.

- A **higher ratio** → larger portion of capital is funded by debt → **higher financial risk** (interest burden, repayment obligations).
- A **lower ratio** → business depends more on equity → **lower long-term risk**, but returns to shareholders may be lower.
- It helps assess the **solvency and financial stability** of a company.
- If Debt to Capital Employed = **0.60**,
→ 60% of the long-term funds come from debt and only 40% from equity
→ High reliance on debt → riskier capital structure.
- If the ratio = **0.25**,
→ Only 25% is debt-funded → safer, more conservative financing.

Suppose a company has:

- Long-term Debt = ₹200 crore
- Capital Employed = ₹500 crore

Debt to Capital Employed = $\frac{200}{500} = 0.40$
Debt to Capital Employed = $\frac{500}{200} = 0.40$

Interpretation:

→ 40% of its long-term funds come from debt.
→ Moderate financial risk.

3. Total Assets to Debt Ratio

Formula:

Total Assets to Debt Ratio = Total Assets / Long-term Debts

Significance: Indicates extent of coverage of long-term debts by assets. Higher ratio indicates assets financed mainly by owners' funds.

Interpretation:

This ratio measures how well the company's total assets **cover its debt obligations**.

Meaning:

- A **higher ratio** → company has more assets compared to debt → **safer solvency position**.
- A **lower ratio** → company has fewer assets to support its debt → **higher long-term financial risk**.

It answers:

“How many rupees of assets does the firm own for every ₹1 of debt?”

If Ratio = 3.5

→ The company owns ₹3.5 of assets for every ₹1 of debt

→ **Strong ability to repay debt**, lower solvency risk.

If Ratio = 1.2

→ Only ₹1.2 of assets for every ₹1 of debt

→ Thin safety margin → **riskier**.

Suppose:

- Total Assets = ₹1,200 crore
- Total Debt = ₹400 crore

Total Assets to Debt Ratio = $\frac{1200}{400} = 3.0$
Total Assets to Debt Ratio = $\frac{400}{1200} = 0.33$

Interpretation:

→ The company has ₹3 in assets for every ₹1 of debt

→ Indicates a **comfortable solvency position**.

The Myth of "Ideal" Ratios

Critical Insight: There is NO universal "ideal" ratio. What's considered healthy varies significantly across industries, business models, and economic conditions.

A “good” or “bad” ratio depends on several factors:

- ✓ **Industry norms**
- ✓ **Company size & business model**
- ✓ **Growth stage of the company**
- ✓ **Economic conditions**
- ✓ **Management strategy**
- ✓ **Competitor performance**

Example:

A current ratio of 1.2 may be excellent for a supermarket (fast inventory turnover) but very poor for a construction company (slow cash cycles).

Key Takeaways for Analysis

✓ **Context is everything**—Industry, business model, and strategy matter more than absolute numbers

- ✓ **Trends matter more than snapshots**—One quarter's ratio means little; 3-year trends tell the story
- ✓ **Compare apples to apples**—Benchmark against direct competitors, not broad industry averages
- ✓ **Look for consistency**—All ratios should tell the same story; contradictions need investigation
- ✓ **Cash is king**—Even with good ratios, poor cash flow is a red flag
- ✓ **Quality of earnings matters**—High profits from one-time gains vs. sustainable operations
- ✓ **Business model determines "normal"**—Amazon's "poor" current ratio is actually brilliant
- ✓ **Think like an owner**—What would you want to see if this were your business?

Practice Questions

Short Answer Questions

1. What do you mean by Ratio Analysis?
2. What are various types of ratios?
3. What relationships are established to study turnover ratios?
4. How is liquidity measured in a business?
5. What is the importance of the average age of inventory?

Long Answer Questions

1. What are liquidity ratios? Discuss the importance of current and liquid ratios.
2. How would you study the solvency position of a firm?
3. What are various profitability ratios and how are they calculated?
4. Compare and contrast current ratio and quick ratio as measures of liquidity.

Case Study: Evaluating the Dabur India

You have recently joined **Dabur India** as a **Financial Manager**. Below are the financial statements for the last two years. Your role is to analyze the firm's financial performance and suggest ways to strengthen its business position.

Questions:

Q1. Compute the following financial ratios for FY 2023 and FY 2024:

- Current Ratio
- Quick Ratio
- Debt-Equity Ratio
- Net Profit Margin
- Return on Capital Employed (ROCE)

Q2. Interpret the overall financial health and profitability.

Q3. What can you infer about liquidity and profitability trends?

Income Statement

Profit & Loss account of Dabur India (in Rs. Cr.)	25-Mar	24-Mar
	12 mths	12 mths
INCOME		
Revenue From Operations [Net]	8,924.85	9,010.58
Total Operating Revenues	9,070.71	9,135.60
Other Income	451.94	417.62
Total Revenue	9,522.65	9,553.22
EXPENSES		
Cost Of Materials Consumed	3,270.74	3,300.25
Purchase Of Stock-In Trade	1,678.41	1,579.05
Changes In Inventories Of FG,WIP And Stock-In Trade	12.9	49.18
Employee Benefit Expenses	776.86	782.14
Finance Costs	99.58	81.14
Depreciation And Amortisation Expenses	250.93	208.86
Other Expenses	1,624.89	1,585.90
Total Expenses	7,714.31	7,586.52
Profit/Loss Before Exceptional, ExtraOrdinary Items And Tax	1,808.34	1,966.70
Total Tax Expenses	405.12	457.49
Profit/Loss For The Period	1,403.22	1,509.21

Balance Sheet

Balance Sheet of Dabur India (in Rs. Cr.)	25-Mar	24-Mar
	12 mths	12 mths
EQUITIES AND LIABILITIES		
SHAREHOLDER'S FUNDS		
Equity Share Capital	177.23	177.2
Reserves and Surplus	7,246.06	6,738.17
Total Shareholders Funds	7,423.29	6,915.37
NON-CURRENT LIABILITIES		
Long Term Borrowings	249.42	498.93
Tax Liabilities	133.25	100.5
Other Long Term Liabilities	74.32	66.94
Long Term Provisions	62.58	60.04
Total Non-Current Liabilities	519.57	726.41
CURRENT LIABILITIES		

Short Term Borrowings	35.38	198.99
Trade Payables	2,341.97	2,058.93
Other Current Liabilities	521.11	471.48
Short Term Provisions	164.15	161.63
Total Current Liabilities	3,062.61	2,891.03
Total Capital And Liabilities	11,005.47	10,532.81
ASSETS		
NON-CURRENT ASSETS		
Tangible Assets	1,943.71	1,810.01
Intangible Assets	48.51	23.8
Capital Work-In-Progress	134.66	161.54
Other Assets	43.4	44.39
Fixed Assets	2,172.50	2,062.43
Non-Current Investments	4,730.46	4,888.30
Long Term Loans And Advances	25.71	38.56
Other Non-Current Assets	146.7	111.15
Total Non-Current Assets	7,075.37	7,100.44
CURRENT ASSETS		
Current Investments	1,628.36	1,260.17
Inventories	1,288.07	1,149.88
Trade Receivables	600.58	719.21
Cash And Cash Equivalents	133.02	71.11
Short Term Loans And Advances	12.85	12.85
Other Current Assets	267.22	219.15
Total Current Assets	3,930.10	3,432.37
Total Assets	11,005.47	10,532.81